Standard Operating Manual

Cello Compound Semiconductor Etcher

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1. Picture and Location

Fig 1 Cello Compound Semiconductor Etcher

This tool is located at NFF Phase II Room 2240.

2. Process Capabilities

2.1 Cleanliness Standard

Cello Compound Semiconductor etcher is classified as a **Non-standard** equipment

2.2 Possible Etching Materials

1. GaN
2. Sapphire
3. III-V semiconductor

2.3 Process Specification

What the Cello Compound Semiconductor CAN do

1. 2 inches full substrates
2. Small samples

What the Cello Compound Semiconductor CANNOT do

1. Substrate larger than 2 inches in diameter
2. Tiny samples that can be blow away.
3. Dirty samples that will contaminate the top plate
4. Metal film etching

3. Contact List and How to Become a User

3.1 Emergency Responses and Communications

In case of emergency issues, please contact NFF staffs,

1. Preason Lee – Deputy Safety office (x7900),
2. CK Wong – senior technician (x7226)

In case of technical help, please contact NFF staffs,

1. CK Wong – senior technician (x7896)
2. Casper Chung –technician (x7896)
3. Brial Kwok – technician (x7896)

3.2 Training to Become a Qualified Cello Compound Semiconductor Etcher User

Please follow the procedures below to become a qualified user of the Cello Compound Semiconductor Etcher:

1. Read all materials provided on the NFF website of the Cello Compound Semiconductor Etcher.
2. E-mail to NFF, requesting Cello Compound Semiconductor Etcher operation training.

4. Operating Procedures

4.1 System Description

The Cello Compound Semiconductor Etcher full load lock ICP system offer anisotropic etching of GaN, Sapphire, III-V compound semiconductor material. The ICP system uses inductive coupled design, so high density plasma can be formed in the process chamber, this in return capable for III-V compound semiconductor material etching. Cl2 and BCl3 gases are used for the etching, and He gas is used for backside cooling to ensure better control of the etch rate, selectivity and profile.

4.2 Safety Warnings

1. If the equipment failure while being used, never try to fix the problem by yourself, please contact NFF etching module staffs
2. In emergency, please push any one of the red emergency buttons (Fig. 2) to interrupt the equipment power, and report to the NFF staffs immediately. DO NOT attempt to resume the equipment on before the problem is solved.

3. During process, if something going wrong and you are not sure what happened, please check and write down the alarm information and report to NFF etching module staffs.

Fig. 2 Emergency button location

4.3 Operation Precautions and Rules

1. Please reserve the time slot on your own, and make sure you use your own time slot to do the etching process
2. Please fill all the details of the log-sheet attached, i.e. date, name, project number, email, project details, material to be etched…
3. Do not operate the equipment unless you are properly trained and approved by NFF staffs
4. Do not leave an on-going etching process unattended
5. Do not skip any cycle purge steps during the process
6. Do not change details of the recipe, if you need to start a new recipe, please

consult NFF etching module staffs

7. Do not use the equipment after office hour

4.4 Initial Status checks

1. Please check the status of Cello Compound Semiconductor Etcher shutdown notice posted in the NFF reservation website
2. According to the reserved time slot, please check-in the equipment on your own
3. After check-in the equipment, please check correct name and project number that printed on the etching module LCD screen (Fig. 3).
4. If you failed to check in the machine, there is an interlock, and you can not operate the machine.
5. Before operate the machine, please make sure you have read and fill the details of the check list.

Fig.3 Reservation and Equipment check-in status LCD

4.5 Initial system checks

- Please check any wafer left in the load lock chamber

- Please check the equipment touch screen is ready to do the etching process (Fig.4)

- Please click on the Process page at the top left corner, and click on the Assistant page at the top right corner. Then, make sure the Vacuum Control is in AUTO mode (green), and confirm the followings:

  1. DP (dry pump) is turned ON (green)

  2. FV (fore line valve) is turned ON (green)

  3. TB (turbo pump) is turned ON (green)

  4. HV(High vacuum gate valve) is turned ON (green)

- Please make sure the default base pressure $5x10^{-5}$ Torr designed in recipe has been achieved, otherwise WAIT!

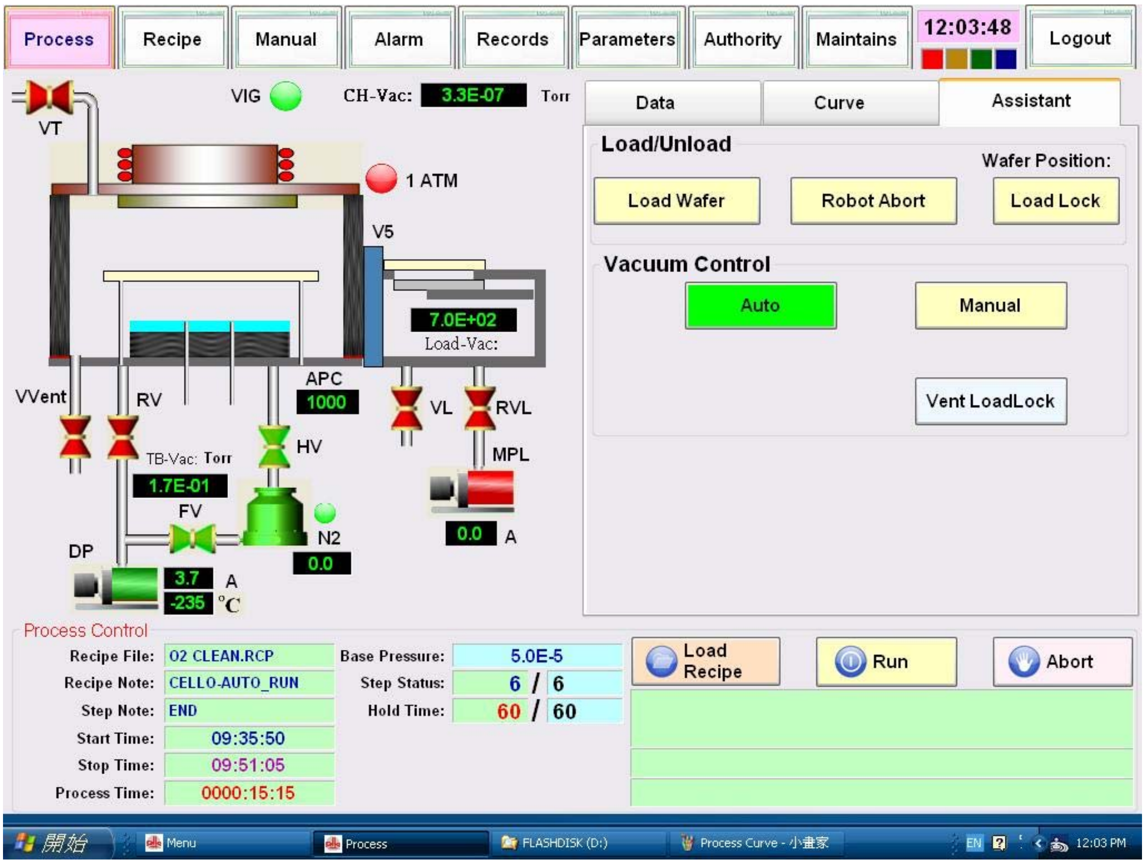

Fig. 4 Process page with vacuum control and process data.

4.6 Preparation before Etching

You are advised to do the following steps before starts to do the etching process.

1. Make sure the shutoff valves of the Cl2 and BCl3 gas delivery system are opened, so that the gas can be delivered to the system, please consult NFF staffs for help.

2. Ask NFF technician to turn the Heater Tie on manually.

4.7 Recipe Selection

Please make sure you select a right recipe, a wrong recipe may get unexpected results. Please read the following Recipe Loading procedures:

1. Click on the Recipe page at the 2nd top left corner (Fig. 5)
2. Click the OPEN button to open the desired recipe.
3. If the recipe you want is not listed in the recipe pull down menu, please consult NFF etching module staffs.
4. Make sure details in the recipe are correct, i.e. process time, power, pressure, etc…, then save it.
5. Go back to the Process page; click Load Recipe to load the desired recipe to the memory, and confirm it by checking the left hand side Recipe File name (Fig. 4)

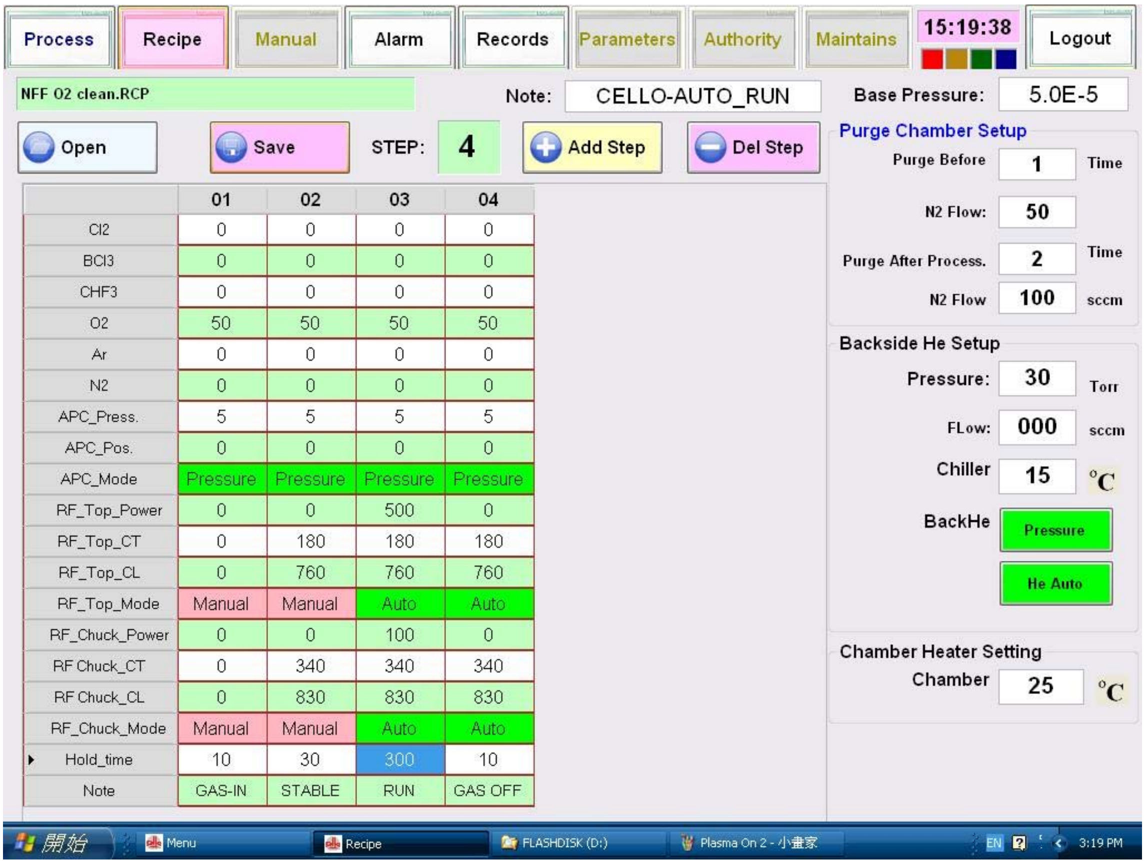

Fig.5 Recipe Page

4.8 Process Run

The whole etching sequence involves,

1) load lock vent, 2) set wafer, 3) pump down and wafer loading, 4) etching process, 5) chamber cycles purge, 6) wafer unloading, 7) load lock cycles purge, 8) load lock vent

1) Load lock chamber vent

- In Process page, under Vacuum Control, click VENT Load Lock button. At the same time, operation mode will be change from AUTO to MANUAL mode (Fig. 4).
- Wait until the Load-lock chamber is displayed 7.6e+2 Torr (about 5 minutes).
- Verify the Load-lock chamber door seal has decompressed.
- The Load-lock chamber door can be open now

2) Set wafer

- Set substrate on the carrier plate and align the plate with the load lock arm (Fig. 6)
- Small samples should be stick on the carrier plate for the etching process

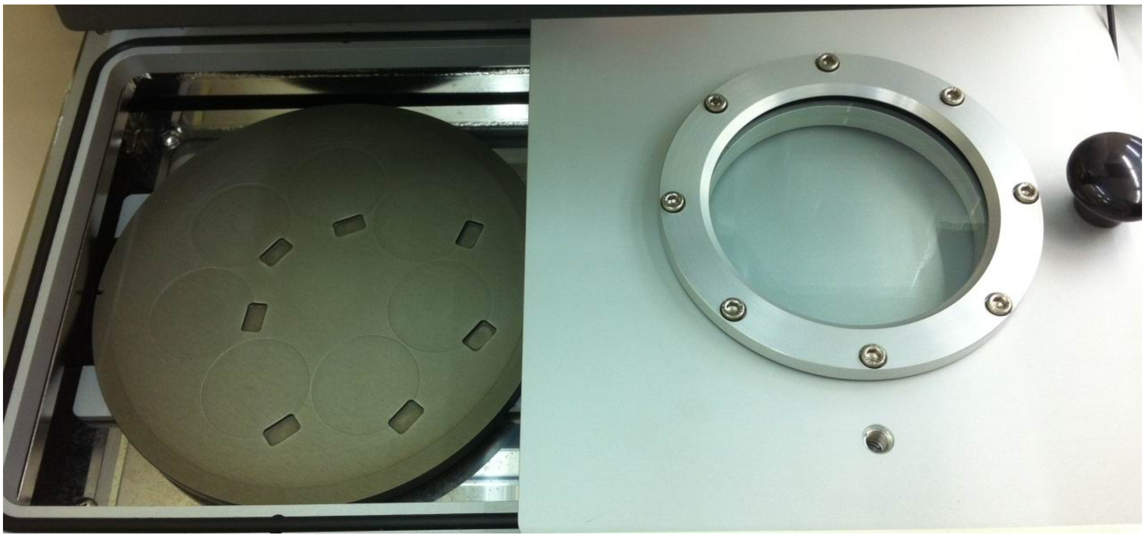

Fig.6 Set substrate

3) Pump down, wafer loading and load recipe

- Close the Load-lock chamber door after set your samples on the plate.
- Click the RUN button in Process page to initiate the etching process (Fig. 7)

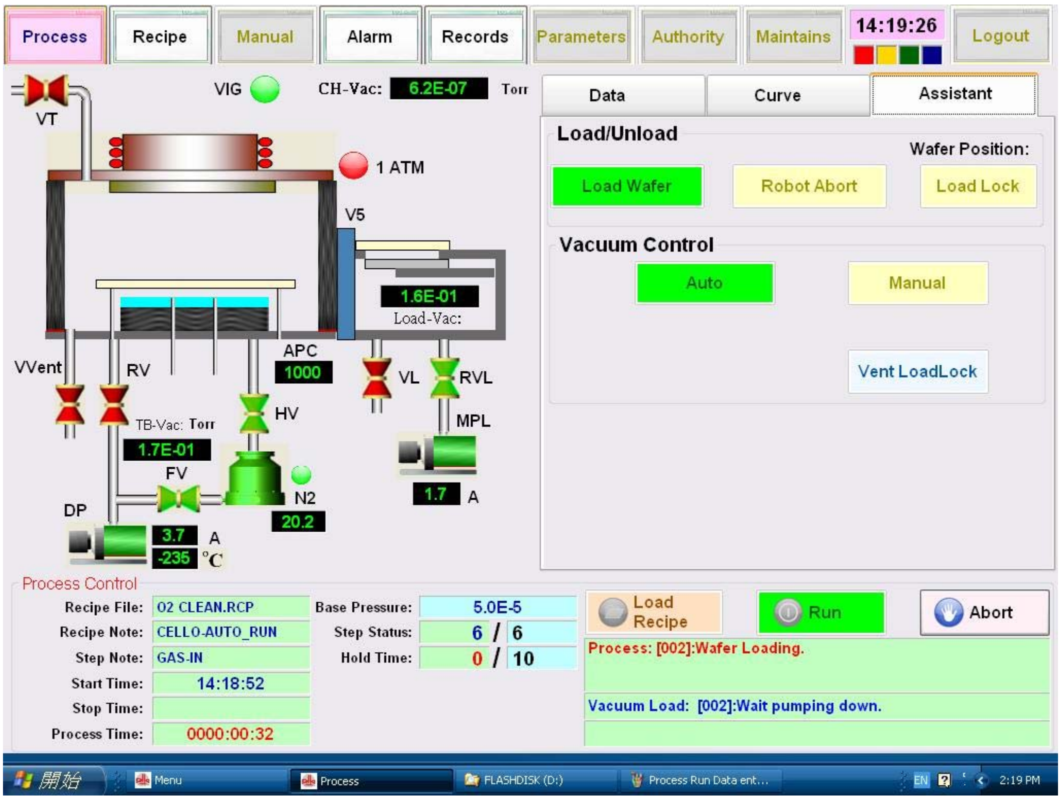

Fig. 7 Run selected recipe

- Then, input your desired log file name in the box come out (Fig.8)

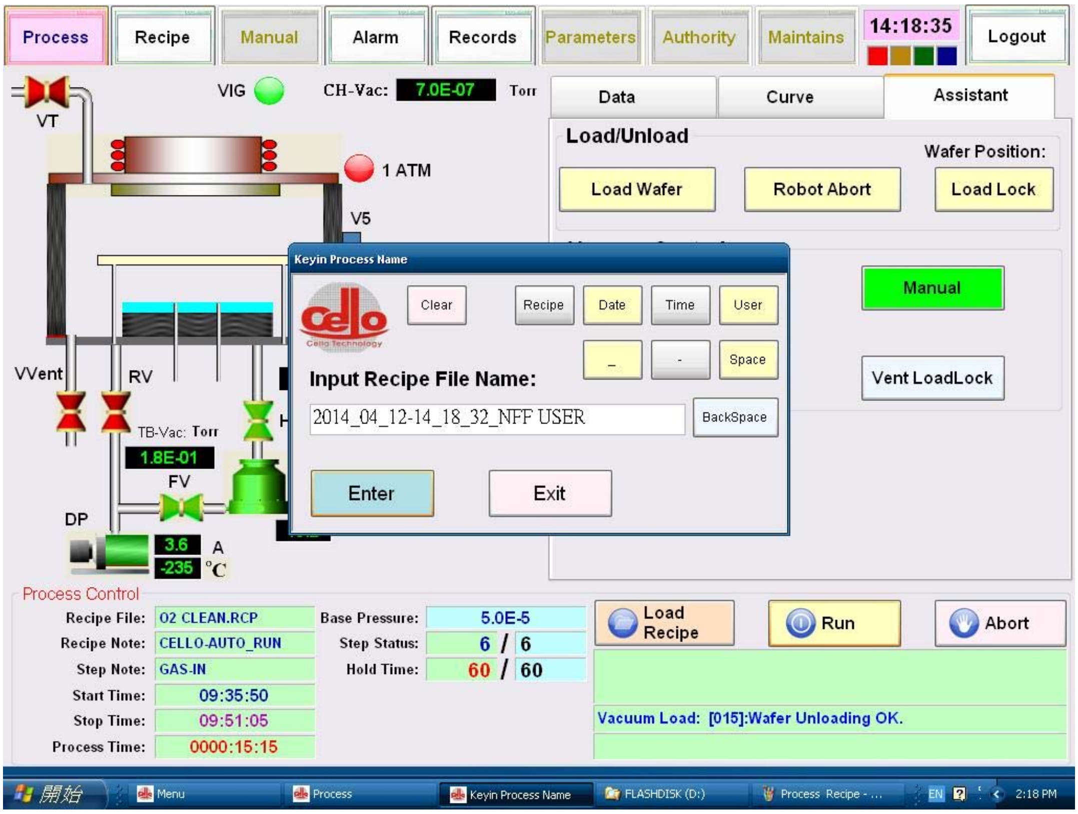

Fig. 8 log file name entry box

- Make sure the load lock chamber door was correctly closed; otherwise the load lock chamber cannot be pump down.
- Check that the roughing pump MPL and roughing valve RVL turn on within a few seconds. (Fig.7)
- Wait a few minutes for the load lock chamber pump down
- When the crossover pressure reached, then load lock arm will send carrier plate into process chamber
- Then, carrier plate sit at the process chamber and clamp down.
- Load lock gate valve closed, and process chamber pump down to base pressure.

4) Start Etching Process

- In the Process page right half, click on the Data page, all the etching data can be observed in this area (Fig. 9).

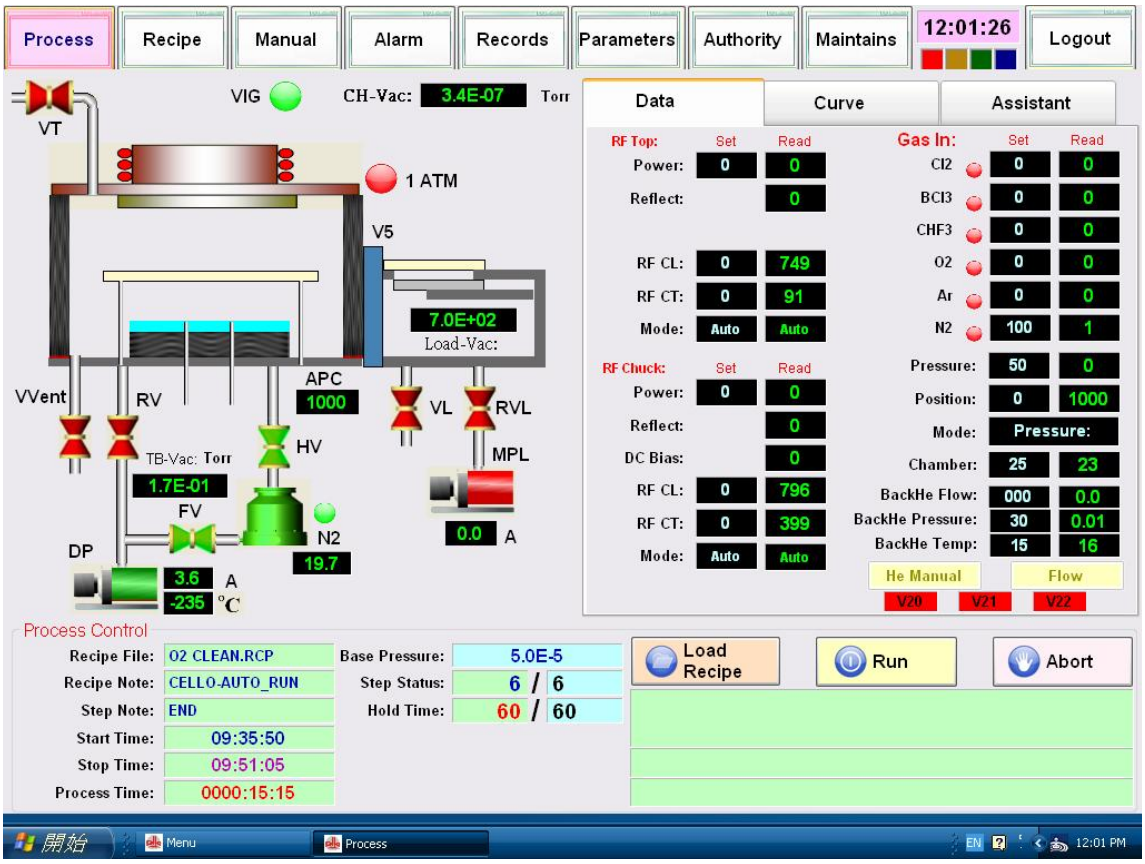

Fig. 9 Process Data page.

- In the Process page low left half, under STEP NOTE row, it shows the step name of the current running step
- When the process chamber base pressure reached, cycle purges steps will be perform, and then Helium leak up rate check
- After that gas start to flow and pressure stabilize
- Next RF power will be given out, and RF matching unit start to match the impedance inside the chamber.
- Effective RF matching can minimize the reflected power, and increase the forward power until it reached the set point, then plasma ignite, and the color of

the chamber shown in the schematic will become “red” (Fig.10)

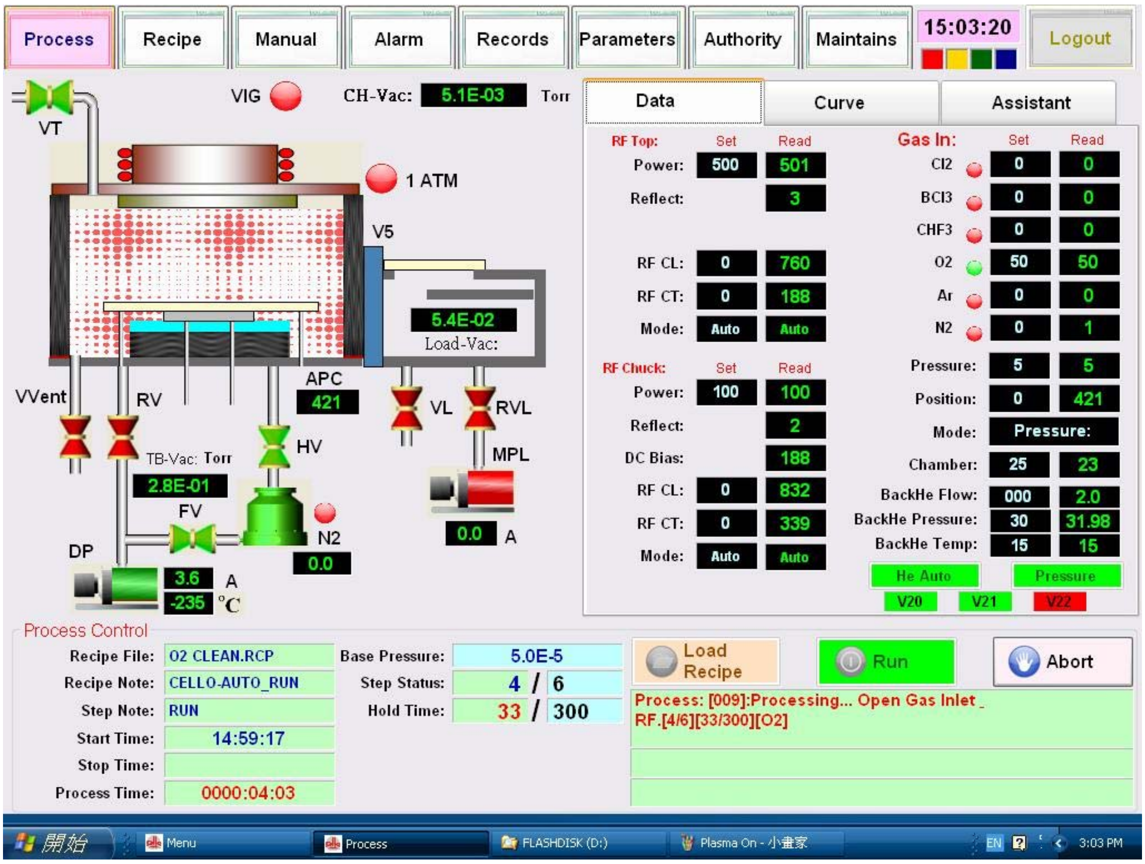

Fig.10 Plasma On

- All the etching parameters can be obtained through the display and “ABORT” function keys can be used to terminate the process during the run.

5) Chamber Cycles Purge

After all the etching steps, a few cycle purge steps will be carry out to purge and pump out all the hazard gases used in the process.

6) Wafer Unloading, 7) Load lock Cycles Purge and 8) Load lock Vent

- After purge cycles, the wafer will be unload back to the load lock chamber automatically.
- Then, load lock chamber will be vent to atmosphere, and there will be a Process

Finished window will be come out (Fig. 11)

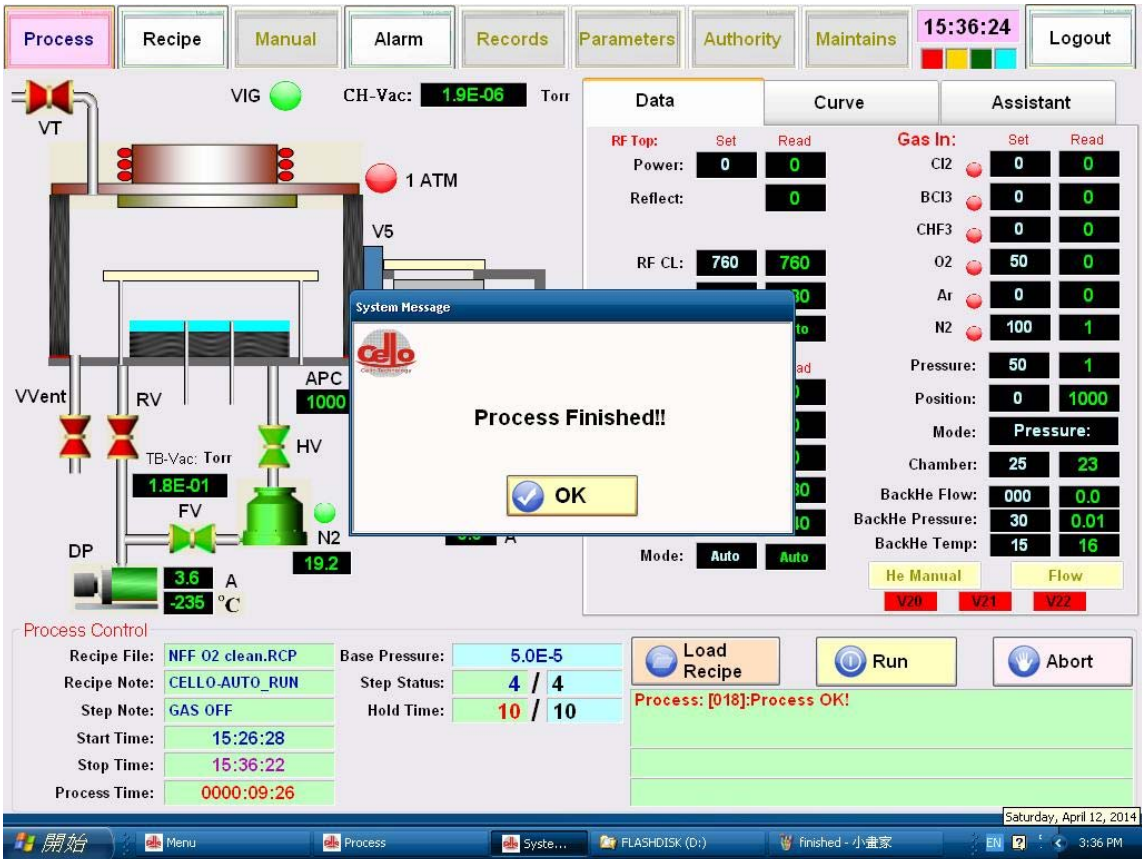

Fig.11 Process Finished

- Open the load lock chamber door and get the samples back
- Ask NFF staff to pump down the load lock chamber.

4.9 Process Recording during the process

1. Please be reminded you are required to fill all the details of the log sheets
2. However, if you fail to do this, a punishment will be given
3. Write down any problems or comments in the log sheets

4.10 Clean up

1. Clean up the area
2. Return items to their proper locations

4.11 Check out

Check out the equipment in the NFF equipment reservation website immediately after use.